Smart Crowd Count

Real-time Crowd Detection using Computer Vision

Project by Jupalli Sanjana

Problem Statement & Objective

Current Challenges

- Manual monitoring is inefficient and error-prone
- CCTV footage remains underutilized
- Absence of real-time analytics capabilities
- Limited automated alert systems

Our Solution

- Detect and count people accurately
- Monitor designated zones continuously
- Provide real-time analytics and alerts
- Enable comprehensive admin control

System Architecture

1. **Camera Input**
   Video feed capture
2. **Frame Processing**
   Image extraction
3. **AI Detection**
   Human recognition
4. **Tracking**
   ID assignment
5. **Database**
   Data storage
6. **Dashboard**
   Visualization
7. **Admin Panel**
   Control center

This end-to-end pipeline transforms raw video into actionable insights through computer vision and real-time processing.

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System Procedure

Step-by-Step Workflow

01

Capture video from camera

Continuous stream acquisition

02

Convert into frames

Image sequence generation

03

Detect humans using AI

YOLO/SSD model inference

04

Assign ID to each person

Unique tracking identifier

05

Count total people

Aggregate detection results

06

Check zone occupancy

Geofencing analysis

07

Compare with threshold

Limit validation check

08

Display status

Normal/Overcrowded alert

09

Send data to dashboard

Real-time visualization

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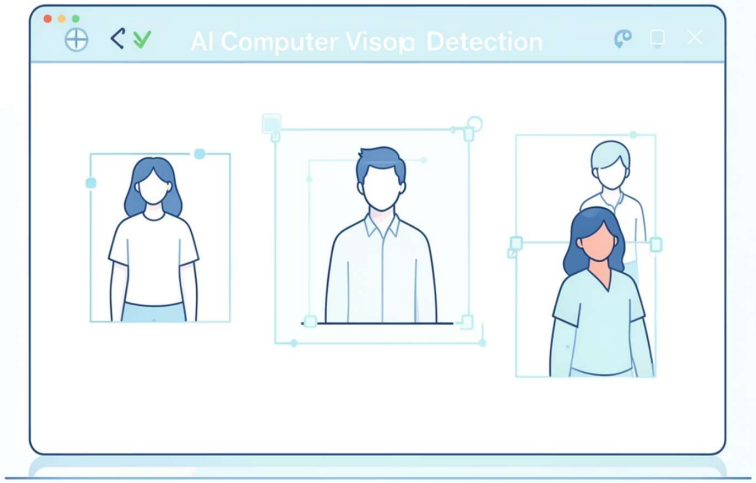
Live Detection Output



Key Metrics Displayed

- **Total People count:** Real-time headcount
- **Status indicator:** Normal/Overcrowded
- **Zone boundary:** Defined monitoring area
- **Zone occupancy:** Current/Maximum (e.g., 1/5)
- **Person ID tracking:** Unique identifiers
- **Real-time updates:** Live refresh

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Zone Monitoring Logic

Define Zone Area

Establish polygon boundaries for monitoring

Set Threshold Limit

Configure maximum capacity (e.g., 5 people)

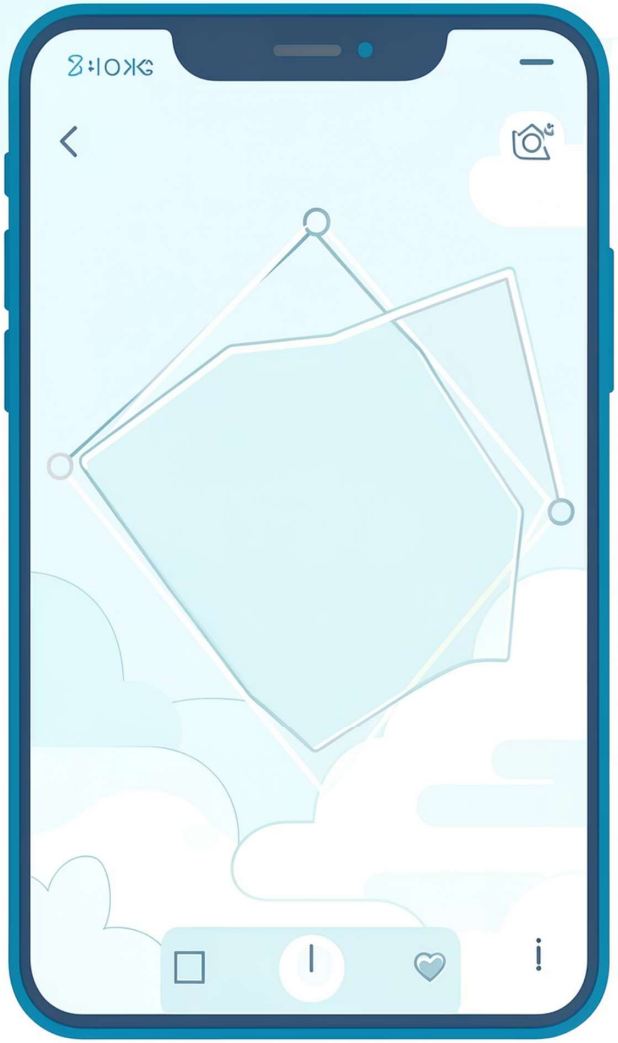
Count Occupancy

Track people inside defined zone

Compare & Alert

Trigger warning if threshold exceeded

Example: Zone occupancy shows 1/5 → Status: Normal



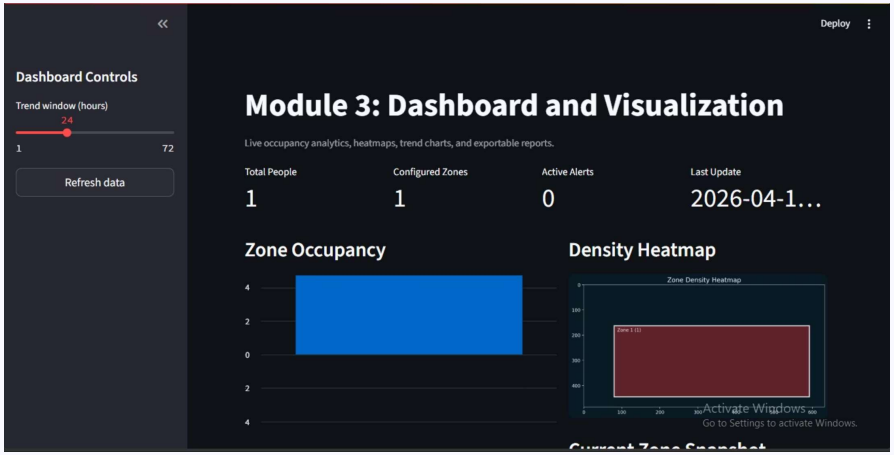
Dashboard Overview & Analytics

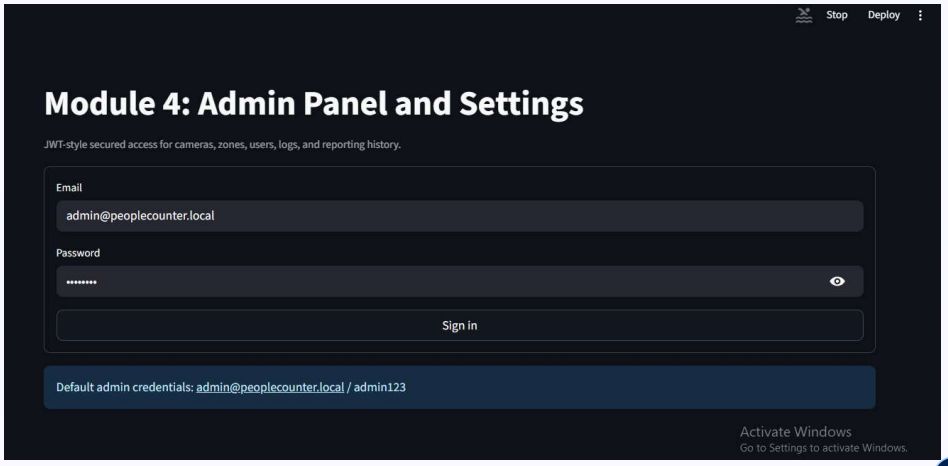
Real-Time Metrics

- **Total People:** Current headcount across all zones
- **Configured Zones:** Active monitoring areas
- **Active Alerts:** Overcrowding notifications

Visual Analytics

- **Zone Occupancy Graph:** Time-series trends
- **Density Heatmap:** Spatial distribution
- **Crowd Trends:** Historical patterns





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Admin Panel & Camera Management

Secure Login

Protected authentication with role-based access control

Camera Registration

Add new cameras with unique identifiers and locations

Source Selection

Configure webcam, IP camera, or video file inputs

Location Setup

Define physical placement and zone boundaries

Activation Control

Enable or disable monitoring for each camera

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Applications, Advantages & Limitations

Applications

- **Smart Cities:** Public space management
- **Surveillance:** Security monitoring
- **Events:** Crowd control
- **Banks:** Queue management
- **Malls:** Traffic flow analysis

Advantages

- **Real-time:** Instant detection
- **Automated:** No manual effort
- **Scalable:** Multiple cameras
- **Cost-effective:** Low hardware needs

Limitations

- **Lighting dependency:** Performance varies
- **Dense crowds:** Occlusion issues
- **Hardware:** GPU requirements
- **Privacy:** Data handling

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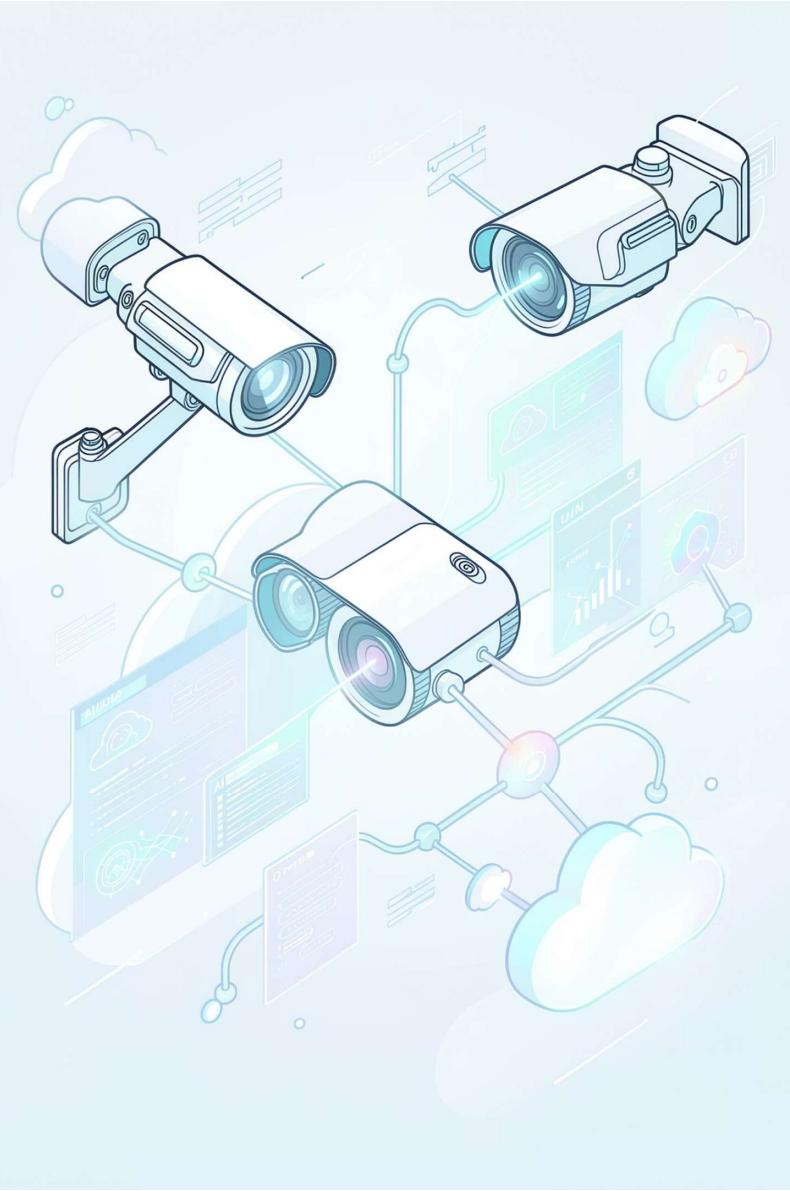
Conclusion & Future Scope

Conclusion

AI-based crowd monitoring significantly enhances traditional surveillance by combining accurate detection, real-time analytics, and centralized control. This system transforms passive CCTV into an intelligent monitoring solution.

Future Enhancements

- **Alert System:** SMS and email notifications
- **Mobile Dashboard:** iOS/Android apps
- **Multi-Camera Integration:** Synchronized monitoring
- **Cloud System:** Scalable infrastructure
- **Behavior Analysis:** Abnormal activity detection



Thank You

Questions?